

Jeevan Thapa

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Scholar: [Google Scholar](#)

Ph.D. candidate specializing in continual learning, parameter-efficient adaptation, and user modeling. First author of papers at ICML and ECIR, with industry research experience developing continual user-modeling methods for recommender systems at Zillow.

RESEARCH FOCUS

Continual Learning and Generalized Category Discovery | Parameter-Efficient Adaptation | Recommender Systems and User Modeling | Gene-Disease Association

EDUCATION

Ph.D. in Computing and Information Sciences

Aug 2022–Feb 2028 (Expected)

Rochester Institute of Technology

Rochester, NY

GPA: 3.97/4.00

Relevant Coursework: Deep Learning, Statistical Machine Learning, Non-Convex Optimization for Modern Machine Learning

Bachelor's Degree in Computer Engineering

Nov 2015–Sep 2019

Pulchowk Campus, Tribhuvan University

Lalitpur, Nepal

Relevant Coursework: Data Mining, Artificial Intelligence, Big Data Analytics, Probability and Statistics

PUBLICATIONS

J. Thapa, R. Li. *Bayesian Adaptation of Network Depth and Width for Continual Learning*. In Proceedings of the 41st International Conference on Machine Learning (**ICML**), 2024.

J. Thapa, S. Zhao, K. Shindo. *Evolving Mixture of Low-Rank Experts for Continual User Modeling*. In Proceedings of the European Conference on Information Retrieval (**ECIR**), 2026.

J. Thapa, R. Li. *Cross-Task Representation Alignment for Exemplar-Free Class-Incremental Learning*. Under review.

J. Thapa, M. Thapa, R. Li. *Probabilistic Graph Propagation with Module Diversity Regularization for Disease Module Discovery*. Under review.

RESEARCH AND INDUSTRY EXPERIENCE

Graduate Research Assistant

Aug 2022 – Present

Rochester Institute of Technology

Rochester, NY

- Built a Bayesian continual learning framework that dynamically adapts network depth and width for evolving tasks; published at **ICML 2024**.
- Developed a cross-task representation-alignment framework that improved average accuracy by 2.16 percentage points for exemplar-free class-incremental learning; manuscript under review.
- Developed a probabilistic graph propagation framework with module diversity regularization that integrates bioinformatics data for gene-disease association prediction.
- Designed a parameter-efficient adaptation method for continual generalized category discovery using full-covariance Gaussian prototypes and evaluating the approach on medical imaging datasets.

Research Scientist Intern

May 2025 – Sep 2025

Zillow Group

Remote

- Developed a mixture-of-experts (MoE) architecture with rank-1, LoRA-style experts for continual user modeling, enabling parameter-efficient adaptation across sequential recommendation and user-attribute prediction tasks; published at **ECIR 2026**.
- Designed a semantic expert-selection strategy that routes each task to relevant prior experts, reducing cross-task interference while reusing transferable knowledge.
- Outperformed the strongest baselines by 2.88% relative HitRate@5 (recommendation) and 1.37% accuracy (attribute prediction).

Machine Learning Engineer

Sep 2019 – Jun 2022

Fusemachines

Kathmandu, Nepal

- Led the development of a multimodal machine-learning pipeline for identifying potential trafficking activity in online advertisements using video, image, and text data.
- Built image-text contrastive models for advertisement matching, face-based identity linking, and BERT-based social-handle extraction, improving cross-ad linkage accuracy by 35%.
- Developed a lightweight object detector that increased inference throughput by 47%, then deployed it on NVIDIA Jetson Nano devices for real-time waste-type and disposal-intent classification.
- Authored computer-vision and time-series instructional materials for the [Fusemachines AI Education Program](#) for AI democratization.

SKILLS

Programming

Python, C++, C

ML Frameworks

PyTorch, TensorFlow, scikit-learn, Transformers, PEFT, Transformers4Rec

Data and Experimentation

NumPy, Pandas, NVTabular, MLflow, Matplotlib, Seaborn

Tools and Platforms

AWS, Git, Linux, Bash, Jira, LaTeX

AWARDS AND HONORS

- **RIT Ph.D. Scholarship** — Full financial support through NSF- and NIH-funded research projects.
- **Tribhuvan University Full Scholarship** — Granted by the Government of Nepal after ranking 11th nationwide in the entrance examination (4% acceptance rate).
- **Fusemachines AI Fellowship** — Awarded a merit-based scholarship through a nationwide selection process to pursue the AI MicroMasters program.

TALKS

- **Poster Presentation** — “Bayesian Adaptation of Network Depth and Width for Continual Learning” at **ICML 2024**, Vienna, Austria.
- **Guest Lecture** — “Continual Test-time Adaptation” and “Deep Learning Toolkit” in graduate course CISC-865: Deep Learning, Rochester Institute of Technology.

ACADEMIC SERVICES

- Conference reviewer for ICLR 2025–2026, ECCV 2026, and NeurIPS 2026.